

Airflow - Conclusion and evaluation

Navigate to airflow/exam directory

Pull in evaluation compose: **wget**

https://dst-de.s3.eu-west-3.amazonaws.com/airflow_fr/eval/docker-compose.yaml

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow_fr/eval/docker-compose.yaml
--2025-07-08 08:51:08-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow_fr/eval/docker-compose.yaml
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 3.5.205.1, 3.5.206.27
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|3.5.205.1|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5680 (5.5K) [binary/octet-stream]
Saving to: 'docker-compose.yaml'

docker-compose.yaml      100%[=====] 5.55K  --.-KB/s  in 0s

2025-07-08 08:51:09 (146 MB/s) - 'docker-compose.yaml' saved [5680/5680]
```

Create all mount-points needed

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ mkdir -p dags logs plugins clean_data raw_files
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ ls
clean_data dags docker-compose.yaml logs plugins raw_files
```

Ensure Airflow can write everywhere

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ sudo chmod -R 777 dags logs plugins clean_data raw_files
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$
```

Seed the .env with the host's UID/GID: **echo -e "AIRFLOW_UID=\$(id -u)\nAIRFLOW_GID=0" > .env**

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ echo -e "AIRFLOW_UID=$(id -u)\nAIRFLOW_GID=0" > .env
```

Initialise the metadata DB

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ docker-compose up airflow-init
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "exam_default" with the default driver
Creating volume "exam_postgres-db-volume" with default driver
Creating exam_redis_1 ... done
Creating exam_postgres_1 ... done
Creating exam_airflow-init_1 ... done
Attaching to exam_airflow-init_1
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sqlalchemy_conn option in [core] has
airflow-init_1 | s been moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The auth_backend option in [api] has bee
airflow-init_1 | n renamed to auth_backends - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:747 DeprecationWarning: The auth_backend option in [api] has bee
airflow-init_1 | n renamed to auth_backends - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:761 FutureWarning: The auth_backends setting in [api] has had ai
airflow-init_1 | rflow.api.auth.backend.session added in the running config, which is needed by the UI. Please update your config before Apache Airflow 3.0.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarning: The sqlalchemy_conn option in [core] has
airflow-init_1 | s been moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/settings.py:194 DeprecationWarning: The sqlalchemy_conn option in [core] has bee
airflow-init_1 | n moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sqlalchemy_conn option in [core] has b
airflow-init_1 | een moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | DB: postgresql+psycopg2://airflow:***@postgres/airflow
airflow-init_1 | Performing upgrade to the metadata database postgresql+psycopg2://airflow:***@postgres/airflow
airflow-init_1 | [2025-07-08T08:56:56.897+0000] {migration.py:216} INFO - Context impl PostgresqlImpl.
airflow-init_1 | [2025-07-08T08:56:56.899+0000] {migration.py:219} INFO - Will assume transactional DDL.
airflow-init_1 | [2025-07-08T08:56:56.902+0000] {migration.py:216} INFO - Context impl PostgresqlImpl.
airflow-init_1 | [2025-07-08T08:56:56.903+0000] {migration.py:219} INFO - Will assume transactional DDL.
airflow-init_1 | INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
airflow-init_1 | INFO [alembic.runtime.migration] Will assume transactional DDL.
airflow-init_1 | INFO [alembic.runtime.migration] Running stamp_revision -> 88344c1d9134
airflow-init_1 | Database migrating done!

airflow-init_1 | Installing collected packages: threadpoolctl, scipy, joblib, scikit-learn
airflow-init_1 | Successfully installed joblib-1.4.2 scikit-learn-1.3.2 scipy-1.10.1 threadpoolctl-3.5.0

airflow-init_1 | [notice] A new release of pip is available: 23.3.2 -> 25.0.1
airflow-init_1 | [notice] To update, run: python -m pip install --upgrade pip
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sqlalchemy_conn option in [core] has
airflow-init_1 | s been moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
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airflow-init_1 | s been moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/settings.py:194 DeprecationWarning: The sqlalchemy_conn option in [core] has bee
airflow-init_1 | n moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sqlalchemy_conn option in [core] has b
airflow-init_1 | een moved to the sqlalchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init_1 | 2.8.1
exam_airflow-init_1 exited with code 0
```

Fetch starter CSV as well as an empty JSON placeholder

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow_avance_fr/eval/data.csv \
> -O clean_data/data.csv
--2025-07-08 08:59:04-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow_avance_fr/eval/data.csv
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 52.95.156.68, 3.5.205.1
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|52.95.156.68|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 450 [binary/octet-stream]
Saving to: 'clean_data/data.csv'

clean_data/data.csv      100%[=====] 450 --.-KB/s  in 0s

2025-07-08 08:59:05 (13.0 MB/s) - 'clean_data/data.csv' saved [450/450]

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ echo '[]' > raw_files/null_file.json
```

Spin everything up in the background

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/exam$ docker-compose up -d
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Pulling dashboard (datascientest/dashapp:1.0.0)...
1.0.0: Pulling from datascientest/dashapp
35807b77a593: Pull complete
764c5de35d8b: Pull complete
48f663c72035: Pull complete
2452ea7f468f: Pull complete
47cd1317e94f: Pull complete
15994746f30b: Pull complete
Digest: sha256:41fc6db4f311df7381bfc2ab09b0713f82896c109cebc982a9b0f219973af57c
Status: Downloaded newer image for datascientest/dashapp:1.0.0
exam_postgres_1 is up-to-date
exam_redis_1 is up-to-date
Creating exam_dashboard_1 ... done
Creating exam_airflow-scheduler_1 ... done
Creating exam_flower_1 ... done
Creating exam_airflow-worker_1 ... done
Creating exam_airflow-webserver_1 ... done
Starting exam_airflow-init_1 ... done
```

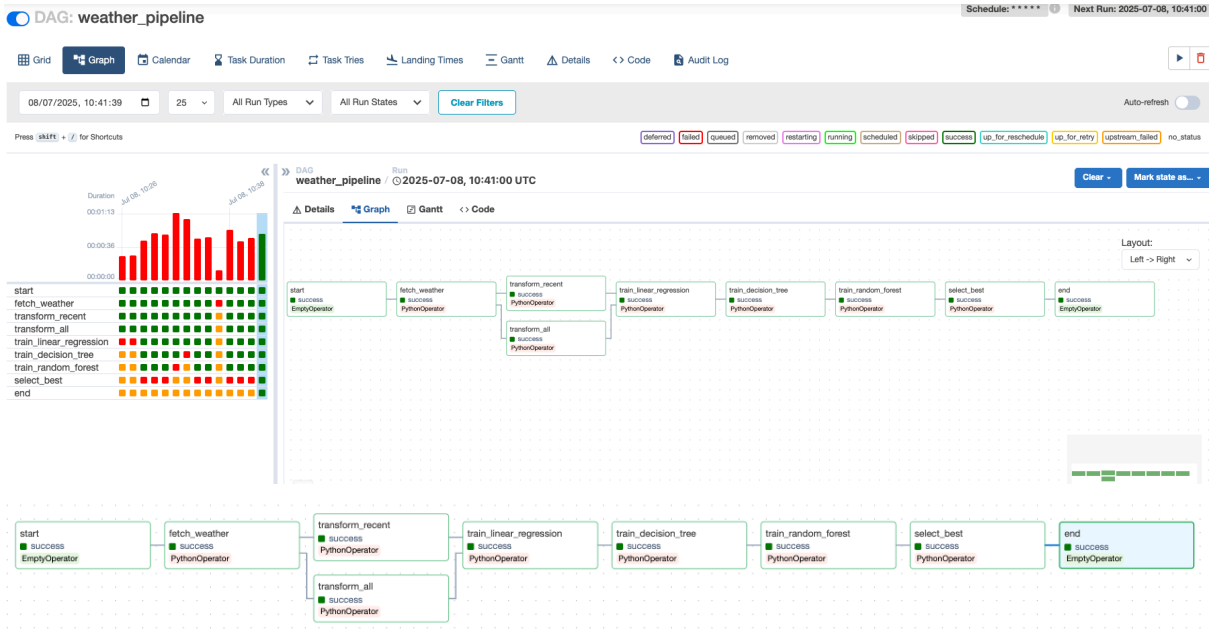
Create dags/weather_pipeline.py file for weather_pipe DAG

Open Airflow UI. Create Airflow variables in *Admin -> Variables*

- Add new record key **cities** and value **["paris","london","berlin"]**
- Add new record key **api_key** and value **openweathermap key**

☐	Key ↑	Val ↑
☐	api_key	*****
☐	cities	["paris","london","berlin"]

Refresh Airflow UI, locate **weather_pipeline** DAG in search filter and trigger it



Open in a URL <http://52.50.160.69:8050/> to verify the dashboard

